Transactions, ACID and Indexing

LumoraAI Presentation Demo • 03 DBMS

ACID

• Atomicity makes a transaction all-or-nothing. Consistency preserves declared integrity constraints. Isolation controls how concurrent transactions observe one another. Durability means committed changes survive failures according to the database's persistence guarantees.

Concurrency

• Locks, timestamps and multi-version approaches help coordinate concurrent transactions. Isolation levels define which phenomena, such as non-repeatable reads or phantom reads, are permitted.

Indexes

• An index is an auxiliary data structure that can reduce the amount of data scanned for selective queries. B-tree indexes support ordered lookups and range access; index maintenance adds storage and write cost.

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